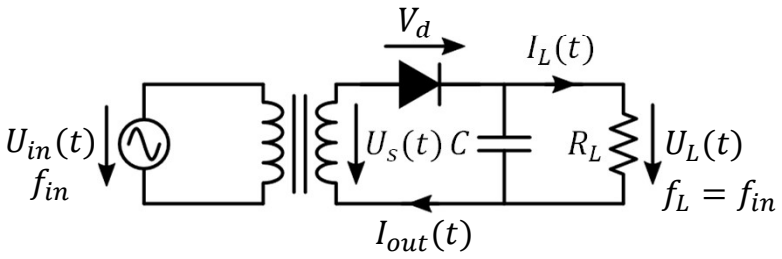
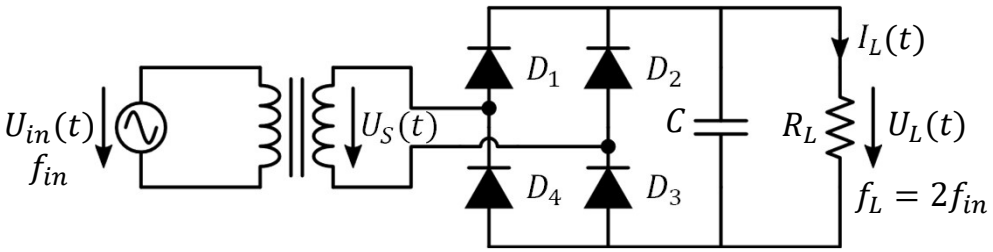
Rectifier + Zener



$$\widehat{U}_L = \widehat{U}_s - V_d$$

$$\Delta T \approx T = \frac{1}{f_{in}}$$

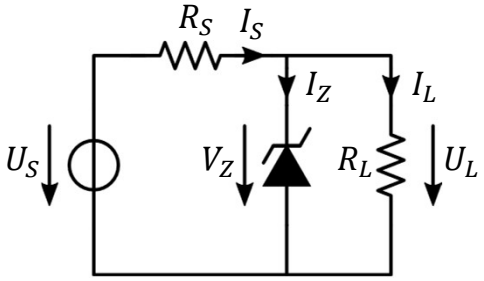


$$\widehat{U}_L = \widehat{U}_s - 2V_d$$

$$\Delta T \approx \frac{T}{2} = \frac{1}{2f_{in}}$$

$$\Delta U_L = \frac{\widehat{U}_L \Delta t}{R_L C}$$

$$\overline{U_L} = \widehat{U}_L - \frac{\Delta U_L}{2}$$



$$I_{Zmax} = I_{Smax} - I_{Lmin} = \frac{U_{Smax} - V_Z}{R_S} - \frac{V_Z}{R_{Lmax}}$$

$$I_{Zmin} = I_{Smin} - I_{Lmax} = \frac{U_{Smin} - V_Z}{R_S} - \frac{V_Z}{R_{Lmin}}$$

$$\Delta I_Z = I_{Zmax} - I_{Zmin} = \frac{\Delta U_S}{R_S} - \Delta I_L$$

$$\Delta I_Z = r_z \Delta I_z$$

R_S protège la diode :

$I_Z < I_{Zmax(lim)}$ $(P_Z < P_{Zmax(lim)})$ quand

$U_S = U_{Smax}$ & $I_L = I_{Lmin}$ $(R_L = R_{Lmax})$

R_S assure le bon fonctionnement de la diode

$I_Z > I_{Zmin(lim)}$ quand

$U_S = U_{Smin}$ & $I_L = I_{Lmax}$ $(R_L = R_{Lmin})$

R_S schützt die Diode:

$I_Z < I_{Zmax(lim)}$ $(P_Z < P_{Zmax(lim)})$ *falls*

$U_S = U_{Smax}$ & $I_L = I_{Lmin}$ $(R_L = R_{Lmax})$

R_S garantiert die korrekte Funktion :

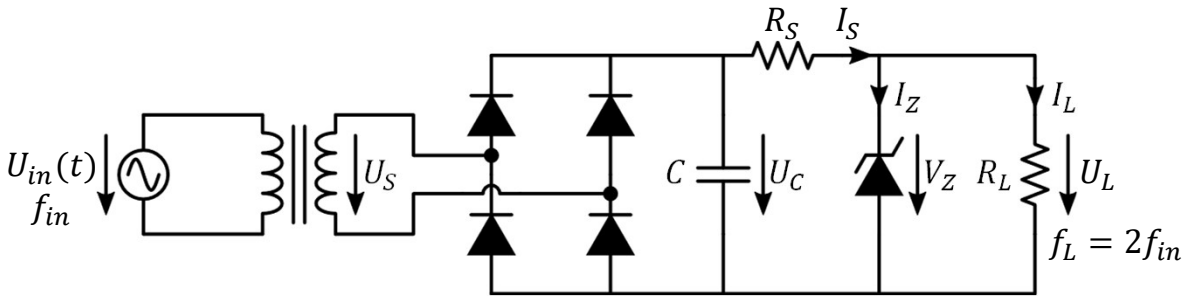
$I_Z > I_{Zmin(lim)}$ *falls*

$U_S = U_{Smin}$ & $I_L = I_{Lmax}$ $(R_L = R_{Lmin})$

$$R_S > \frac{U_{Smax} - V_Z}{I_{Zmax(lim)} + I_{Lmin}}$$

$$R_S < \frac{U_{Smin} - V_Z}{I_{Zmin(lim)} + I_{Lmax}}$$

Rectifier + Zener



$$\widehat{U}_C = \widehat{U}_s - 2\,V_d$$

$$\Delta T \approx \frac{T}{2} = \frac{1}{2f_{in}}$$

$$\Delta U_C = \frac{\hat{I}_S \Delta T}{C} = \frac{(\widehat{U}_C - V_Z)\Delta T}{R_S C}$$

$$\overline{U_C} = \widehat{U}_C - \frac{\Delta U_C}{2}$$

$$\Delta U_L = \frac{\Delta U_C\, r_z}{R_S}$$

$$I_{Zmax} = I_{Smax} - I_{Lmin} = \frac{\widehat{U}_C - V_Z}{R_S} - \frac{V_Z}{R_{Lmax}}$$

$$I_{Zmin} = I_{Smin} - I_{Lmax} = \frac{\widehat{U}_C - \Delta U_C - V_Z}{R_S} - \frac{V_Z}{R_{Lmin}}$$

$$R_S > \frac{\widehat{U}_C - V_Z}{I_{Zmax(lim)} + I_{Lmin}}$$

$$R_S < \frac{\widehat{U}_C - \Delta U_C - V_Z}{I_{Zmin(lim)} + I_{Lmax}}$$